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TD : Ansible

Année universitaire : 2024-2025

Vérification du ping vers la machine B :

```
root@debian:~# ansible 192.168.10.2 -m ping
[WARNING]: Invalid characters were found in group names but not
see details
192.168.10.2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Vérification du ping global :

```
root@debian:~# ansible global -m ping
[WARNING]: Invalid characters were found in group names but not
see details
192.168.10.2 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
192.168.10.1 | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Le module ping permet de savoir si on peut se connecter sur la machine distante ainsi de vérifier si Python est bien présent sur la machine distante.

Machines uptime :

```
root@debian:~# ansible global -m shell -a "uptime"
[WARNING]: Invalid characters were found in group names but not
see details
192.168.10.1 | CHANGED | rc=0 >>
 15:10:07 up 32 min,  2 users,  load average: 0,21, 0,20, 0,14
192.168.10.2 | CHANGED | rc=0 >>
 15:10:07 up 23 min,  2 users,  load average: 0,25, 0,19, 0,13
```

Contenu des groupes :

```
root@debian:~# ansible global --list-hosts
[WARNING]: Invalid characters were found in group names but not
see details
  hosts (2):
    192.168.10.1
    192.168.10.2
root@debian:~# ansible production-server --list-hosts
[WARNING]: Invalid characters were found in group names but not
see details
  hosts (1):
    192.168.10.2
```

Informations des machines :

```
root@debian:~# ansible production-server -m setup
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to
see details
192.168.10.2 | SUCCESS => {
    "ansible_facts": {
        "ansible_all_ipv4_addresses": [
            "192.168.10.2"
        ],
        "ansible_all_ipv6_addresses": [
            "fe80::20c:29ff:fe5f:6419"
        ],
        "ansible_apparmor": {
            "status": "enabled"
        },
        "ansible_architecture": "x86_64",
        "ansible_bios_date": "11/12/2020",
        "ansible_bios_vendor": "Phoenix Technologies LTD",
        "ansible_bios_version": "6.00",
        "ansible_board_asset_tag": "NA",
        "ansible_board_name": "440BX Desktop Reference Platform",
        "ansible_board_serial": "None",
        "ansible_board_vendor": "Intel Corporation",
        "ansible_board_version": "None",
        "ansible_chassis_asset_tag": "No Asset Tag",
        "ansible_chassis_serial": "None",
        "ansible_chassis_vendor": "No Enclosure",
        "ansible_chassis_version": "N/A",
        "ansible_cmdline": {
            "BOOT_IMAGE": "/boot/vmlinuz-6.1.0-12-amd64",
            "quiet": true,
            "ro": true,
            "root": "UUID=1522103f-caf7-473d-8be8-9bee144bb500"
        },
        "ansible_date_time": {
            "date": "2025-03-10",
            "day": "10",
            "epoch": "1741615948",
            "epoch_int": "1741615948",
```

Exercices :

a) Ajout de l'utilisateut toto :

ansible production-server -m ansible.builtin.user a « name : toto home-create: yes »
ansible production-server -m ansible.builtin.group -a « name : group1 »
ansible production-server -m ansible.builtin.group -a « name : group2»
ansible.production-server -m ansible.builtin.user -a « name :: toto group : group1 »
ansible production-server -m ansible.builtin.file -a « src : /*tmp/*myconf state : touch mode : 600 »

ansible.production-server -m ansible.builtin.apt -a « name htop »
ansible.productio-server -m ansible.builtin.service -a « name : cron state : stopped »

b) Création du playbook :

Créer un playbook dans le dossier où il y a l'inventaire :

nano /etc/ansible/playbook.yml

```yaml
---
- hosts: 192.168.10.2
#- remote-user: root

  tasks:
  - name: Creer un utilisateur toto
    ansible.builtin.user:
     name: toto
     create_home: yes


  - name: Creer le groupe1
    ansible.builtin.group:
     name: group1

  - name: Creer le groupe2
    ansible.builtin.group:
     name: group2

  - name: Ajout toto au groupe1
    ansible.builtin.user:
     name: toto
     group: group1

  - name: Creer le fichier /tmp/my-conf avec droits 600
    ansible.builtin.file:
     path: /tmp/my-conf
     state: touch
     mode: '600'

  - name: Installer le programme htop
    ansible.builtin.apt:
     name: htop

   - name: Arreter le service crond
     ansible.builtin.service:
      name: cron
      state: stopped
```

On lance le playbook : ansible-playbook /etc/ansible/playbook.yml

```
root@debian:~# ansible-playbook /etc/ansible/playbook.yml
[WARNING]: Invalid characters were found in group names but not replaced, use -vvvv to
see details

PLAY [192.168.10.2] ************************************************************

TASK [Gathering Facts] *********************************************************
ok: [192.168.10.2]

TASK [Creer un utilisateur toto] ***********************************************
ok: [192.168.10.2]

TASK [Creer le groupe1] ********************************************************
ok: [192.168.10.2]

TASK [Creer le groupe2] ********************************************************
ok: [192.168.10.2]

TASK [Ajout toto au groupe1] ***************************************************
ok: [192.168.10.2]

TASK [Creer le fichier /tmp/my-conf avec droits 600] ***********************************
changed: [192.168.10.2]

TASK [Installer le programme htop] *********************************************
ok: [192.168.10.2]

TASK [Arreter le service crond] ************************************************
changed: [192.168.10.2]

PLAY RECAP *********************************************************************
192.168.10.2               : ok=8    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

c) Déploiement du serveur apache

On créer un nouveau playbook pour le déploiement du serveur web.

```yaml
---
- hosts: production-server
#- remote-user: root

  tasks:
  - name: Installer apache
    ansible.builtin.apt:
     name: apache2


  - name: Deployer le serveur web apache
    ansible.builtin.service:
      name: apache2
      enabled: yes

  - name: Copier le fichier httpd.conf
    ansible.builtin.fetch:
       src: /etc/apache2/apache2.conf
       dest: ~/
```

Vérification du fonctionnement du playbook :

```
root@debian:~# ansible-playbook /etc/ansible/serveur.yml
[WARNING]: Invalid characters were found in group names but not repla
-vvvv to see details

PLAY [production-server] ********************************************

TASK [Gathering Facts] **********************************************
ok: [192.168.10.2]

TASK [Installer apache] *********************************************
ok: [192.168.10.2]

TASK [Deployer le serveur web apache] *******************************
ok: [192.168.10.2]

TASK [Copier le fichier httpd.conf] *********************************
ok: [192.168.10.2]

PLAY RECAP **********************************************************
192.168.10.2               : ok=4    changed=0    unreachable=0    fa
kipped=0    rescued=0    ignored=0
```